SANDEEP YADAV

Gorakhpur, Uttar Pradesh
+91 7897615826 sandeep7897615826@gmail.com LinkedIn Github

Education

National Institute of Technology (NIT), Hamirpur **Nov 2022- May 2026**
Bachelor of Technology in Computer Science and Engineering ***CGPA : 8.03/10***

Technical Skills

- **Languages :** C, C++, Python, SQL
- **Frame-work/Technologies :** ScikitLearn, NLTK, Numpy, Pandas, Matplotlib, Tensorflow
- **Technical Skills** : SQL, OpenCV, REST API
- **Software & Tools :** GIT, MySQL, VSCode, Jupyter Notebook, Google Collab, Kaggle, Databricks

Relevant Coursework:

- Operating Systems, Data Structures, Algorithms, Database Management Systems, Oops

Projects

Project A: AI For Sales And Advertising

Tools/Technologies: Jupyter Notebook, Python(Numpy, Matplotlib)

- **Comprehensive pricing analysis**: Spearheaded pricing analysis for an e-commerce platform, resulting in a strategic shift that increased premium subscription conversions by **40%**.
- **Thompson Sampling Optimization**: Implemented a rigorous comparison of Thompson Sampling against random selection, revealing actionable insights that led to improved user engagement.
- **Marketing strategy visualizations**: Developed visual representations of strategy selections through histograms, aiding stakeholders in making informed decisions across 10 advertising channels.

Project B: Cricket Match and Fantasy Prediction using Machine Learning

Tools/Technologies: Python (pandas, scikit-learn, OneHotEncoder, RandomForest), Flask, Streamlit, Git, GitHub

- **Data Preprocessing and Feature Engineering:** Cleaned and encoded Cricket match data using OneHotEncoder and assigned weighted importance to matches based on recency and impact.
- **Multi-Model Training**: Trained three separate Random Forest models for winning probability (Team A and B) and match outcome prediction using sample weights.
- **Custom Prediction API**: Built a Flask-based API to accept ground and team input, returning real-time probabilities for each team.
- **Use Case**: Designed for fantasy app players to make informed decisions and cricket enthusiasts to analyze match scenarios using Machine Learning insights.

Project C: Simultaneous Fruit Detection and Diameter Estimation using Multitask DNNs

Tools/Technologies: Python, PyTorch, YOLOv5, Fast R-CNN, OpenCV, NumPy, Pandas, Matplotlib, RGB-D Processing

- **Multimodal Data Fusion:** Engineered a training dataset pairing **11,654 field fruit metrics** across three apple cultivars (*Gala Redlum, Golden Clone B, Granny Smith*) with **1,260 meteorological records** (temperature/humidity).
- **Dual-Branch AI Pipeline:** Combined **YOLOv5** for rapid, multi-scale fruit counting with an amodal **Fast R-CNN architecture** that mathematically reconstructs hidden fruit geometry behind leaf occlusions.
- **Precision Metric Sizing:** Implemented the Pinhole Camera Model on 4-channel RGB-D data and developed a **depth percentile-shifting filter** to eliminate foreground leaf noise, achieving a sizing Mean Absolute Error (MAE) of **under 3.0 mm**.
- **Predictive Growth Modeling:** Built and tuned an **Extra Trees Regressor** to map environmental microclimates onto physiological fruit growth stages, achieving high statistical correlations (R^2 of **0.95 for Gala Redlum**, **0.94 for Golden Clone B**, and **0.93 for Granny Smith**).
- **Quality Control Architecture:** Formulated an automated visibility ratio validation function to programmatically filter out high-uncertainty targets (<70% visibility), ensuring clean, stable longitudinal yield forecasting curves.

Achievements

- Solved **300**+ questions across multiple platforms.
- Achieved 3rd position in HACK 5.0 Hackathon organized by CSEC, NIT Hamirpur. Certificate
- Winner at **Codequest** coding competition at NITH with participants more than 1000.

Certifications

•**Machine Learning A-Z** Certificate